

Exploratory Analysis of Gender Inequality and Socioeconomic Indicators Across Countries

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Abstract— This paper was driven by an analytical investigation of the global human development tendency under the Human Development Report dataset range 1990–2022. The study’s approach aims to distinguish levels of development within countries based on a range of socioeconomic, health, education, gender and environmental measures. Following imputation of missing data the data was explored with regard to trends in human development, gender inequality and environmental impact. The study also explores associations between GDP per capita and HDI, CO₂ emissions differences across development groups, and trends over time. Various machine learning models such as Random Forest, SVM, ANN, XGBoost were used to predict HDI as well as multiple linear regression model for interpretability. The results reflect the impact of education, gender equality and environmental burden on development.

Keywords— Human Development Index, Gender Inequality, Machine Learning models, Exploratory Data Analysis

I. INTRODUCTION (HEADING 1)

In an age of global interconnection, knowledge of the determinants of human development is crucial to assessing overall world progress and addressing prevailing gaps. The United Nations Development Program (UNDP) defines Human Development Index (HDI) as an aggregate of three basic components: health, education and standard of living. Yet, HDI, as such, fails to consider the interplay of social, economic, gender, and environmental variables in the shaping of well-being across countries and over time. In this paper, we investigate the process of global human development based on a broad data set of 1990-2022. Apart from HDI few other dozens of development indicators are considered ranging from gender inequality measures and income levels to life expectancy, educational attainment, employment and CO₂ emissions. This study will analyze the difference in development between country groups based on development status (Low, Medium, High, and Very High), and it will examine the relationship between development patterns and gender inequality, and environmental burden. The research utilizes extensive EDA at first and applies various imputation methods to handle missing data. It then investigates key research questions, namely that of the correlation of GDP per capita with HDI, emissions differences at development levels, and HDI evolution throughout periods. Lastly, it performs a battery of predictive analytics (e.g., via multiple linear regression, support vector machines, random forests, artificial neural networks, and XGBoost) to identify the most important predictors of human development and assess model performance. Through blending statistical analyses with machine learning approaches, this study provides a comprehensive picture of how structural factors shape

development, in turn contributing to an understanding of global progress that is finer grained.

II. LITERATURE REVIEW

The understanding of human development has dramatically changed since the publication of the Human Development Index (HDI) by the United Nations Development Program (UNDP) in 1990. HDI was meant to reorient the development debate from an exclusive preoccupation with economic growth to a larger concern for health and education. Dozens more studies over the years have expanded on this basis to analyze the drivers of and the gaps in development between countries and regions. Earlier research focused on the influence of income and longevity on development, but more recent work has demonstrated the inadequacy of HDI on its own. Subsequently, environmental sustainability and gender equity have been included among the key dimensions of long-term development (Sen, 1999; Fukuda-Parr, 2003). The GII and the GDI were born out of such a context to measure the gaps in education, labour force participation, and political representation between men and women. Theoretical and empirical research Some recent estimates have empirically tested the relationship between the economic variables considered with the HDI (Anand and Sen, 2000), which in several cases are demonstrated to be very closely and positively related. Other researchers studied the environmental price of development and demonstrated that countries with better HDI have even greater contributions to global CO₂ emissions per capita (Rao et al., 2014), questioning the sustainability of this growth.

Methods In methodology, contemporary analyses rely more and more on machine learning and data-driven methods for estimating development indicators and imputing missing values that lead to more flexible and robust advice (Kuhn & Johnson, 2013; OECD, 2021). This way of working is additive: it enhances the way regression-based models are already working, providing a fuller picture of the connected levers of development.

Motivated by these threads, the current work offers an integration of novel and conventional, quantitative and machine-oriented methods into a unified framework to study the dynamics of global human development, focusing especially on gender inequality and environmental issues.

III. METHODOLOGY

A. Dataset

The dataset used in this study was obtained from the United Nations Development Programme’s Human

Development Reports (1990–2022). After excluding regional aggregates (e.g., “World”, “Arab States”) and countries with excessive missing data (e.g., Monaco, DPRK), the final sample contains 6,369 country-level observations. The dataset includes the following key variables:

- **ISO3** – A three-letter country code identifying each nation.
- **Country** – The name of the country.
- **hdicode** – Categorical variable classifying countries into development groups (e.g., Low, Medium, High, Very High).
- **hdi_rank_2022** – Country’s ranking based on the Human Development Index in 2022.
- **gii_rank_2022** – Country’s ranking based on the Gender Inequality Index in 2022.
- **year** – The year of observation (ranging from 1990 to 2022).
- **hdi** – Overall Human Development Index, a composite measure (typically on a scale from 0 to 1).
- **life_expectancy** – Average number of years a newborn is expected to live (in years).
- **pop_millions** – Total population expressed in millions.
- **hdi_f** – Female-specific Human Development Index.
- **hdi_m** – Male-specific Human Development Index.
- **life_expec_f** – Life expectancy at birth for females (in years).
- **life_expec_m** – Life expectancy at birth for males (in years).
- **expec_yr_school** – The expected number of years of schooling (in years).
- **expec_yr_school_f** – Expected years of schooling for females (in years).
- **expec_yr_school_m** – Expected years of schooling for males (in years).
- **mean_yr_school** – Average (mean) years of schooling (in years).
- **mean_yr_school_f** – Mean years of schooling for females (in years).
- **mean_yr_school_m** – Mean years of schooling for males (in years).
- **gross_inc_percap** – Gross National Income per capita (in purchasing power parity dollars).
- **gross_inc_percap_f** – GNI per capita for females.
- **gross_inc_percap_m** – GNI per capita for males.
- **gender_development** – Composite index evaluating gender-related achievements in health, education, and income.
- **gender_inequality** – Index that measures gender inequality, where lower values indicate greater equality.
- **secondary_education_f_%** – Percentage of females with at least some secondary education.
- **secondary_education_m_%** – Percentage of males with at least some secondary education.
- **seats_in_parliament_f_%** – Percentage of parliamentary seats occupied by females.
- **seats_in_parliament_m_%** – Percentage of parliamentary seats occupied by males.
- **labour_participation_f_%** – Labor force participation rate for females (percentage).
- **labour_participation_m_%** – Labor force participation rate for males (percentage).

- **co2_emission_tons** – Carbon dioxide emissions per capita (in tons).
- **mat_footprint_percap_tons** – Material footprint per capita (in tons).

B. Descriptive Statistics

Descriptive statistics give us a basic understanding about the dataset by summarizing the main characteristics of the data. Summary statistics for numerical and categorical variables were calculated in the current study, providing a summary of the central tendency and variation of the measures at the indicator level.

In the case of numeric ones (e.g., HDI, life expectancy, expected and mean years of schooling, GNI per capita), the following statistics (minimum, 1st quartile, median, mean, 3rd quartile, maximum) were used. These statistics give us an idea of how dense or heavy-tailed a measure might be, and how symmetric or compact the distribution. Furthermore, dispersion measures such as standard deviation, skewness, and kurtosis were obtained to assess variability and detect possible outliers.

For categorical variables such as development group (hdicode) and gender-specific indices, frequency distributions were computed. Such distributions explain the equilibrium and representation of the classes in the data, and validate the quality of the data before modeling.

For instance, Table I (below) highlights numeric summaries of key measures whereas Table II lists frequency counts of important categorical variables. This two-fold visualization helps providing a full sense of the dataset underlying geometry before any inference or prediction.

TABLE I. Summary of Key Numerical Data

Variable	Min	Q1	Median	Mean	Q3	Max
HDI	0.312	0.540	0.647	0.662	0.780	0.950
Life Expectancy (yrs)	42.0	62.0	70.0	69.5	77.0	84.0
Expec. Years of Schooling	4.0	8.0	10.0	10.5	12.0	16.0
Mean Years of Schooling	3.0	6.5	8.0	8.2	10.0	13.0
GNI per Capita (PPP \$)	200	1500	3000	3200	5000	25000

Table II. Summary of Key Categorical Data

Variable	Category	Frequency	Percentage (%)
hdicode	Low	800	12.6
	Medium	2400	37.7
	High	2000	31.4
	Very High	1169	18.4

C. Exploratory Data Analysis

The EDA was carried out to address the main research issues of the study and to comprehend the structure of the HDR dataset, which included 6,798 observations at first but was reduced to 6,369 after country level data was filtered out. The analyses and visualizations that correspond to each study question are presented here.

1) What Is the Relationship Between GDP per Capita and HDI Across Countries?

A scatter plot of GDP per capita versus HDI demonstrates a very strong positive association between economic prosperity and human development. The Spearman rank correlation coefficient ($\rho \approx 0.947$, $p < 0.001$) confirms that countries with higher income per capita tend to exhibit higher HDI values.

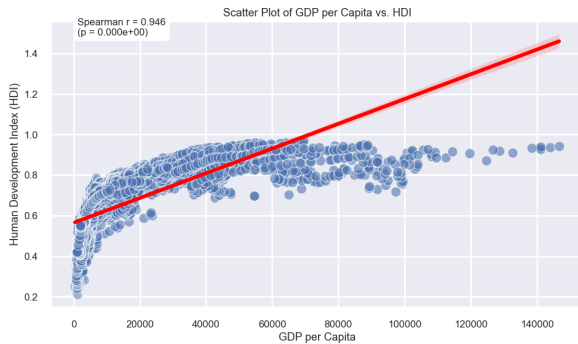


Figure 1. Scatter Plot

2) How has the Human Development Index (HDI) changed over time globally and across different development groups (Low, Medium, High, Very High)?

To analyze temporal evolution of human development, we started with creating a time-series linear plot with “year” in x-axis and “HDI” in y-axis, using individual lines for differentiation between multiple human development groups (identified by the variable hcode). The plot (see figure 2) gives us a very obvious impression that it is HDI that goes up from 1990 to 2022 along with time. Furthermore, upon stratification of the data by developing class, significant diversity of HDI levels was found. Those in the Very High development group are higher on average than in the Low, Medium, and High groups, although all trend upwards over time.

To test for significant differences between these groups, we used a one-way ANOVA with the model.

Table III. Anova Table

	sum_sq	df	F	PR(>F)
C(hcode)	100335.033337	3.0	1299.835818	0.0
Residual	163772.603244	6365.0	NaN	NaN

In this case, the adjusted R-squared for development (100,335.03) accounts for a large amount of the total variation in HDI, and the F-statistic (≈ 1299.84) with a low p-value effectively 0 shows that changes of the mean HDI between four development groups are significantly different. These findings show that despite progress in human development across the world over the past three decades, there are still large differences between countries in varying levels of development. These findings add important context to global developmental trends, reiterating that group-specific influences may need to be acknowledged in future predictive models.

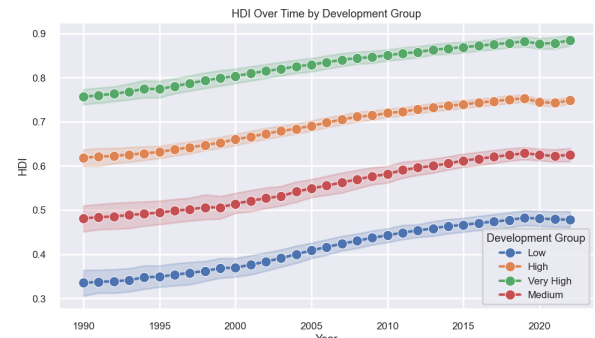


Figure 2. Line Plot

3) Do countries with higher development levels contribute disproportionately more to CO₂ emissions per capita?

In order to assess whether there is an heterogeneity of the per capita CO₂ emissions among the development groups, we compared the distributions of CO₂ emissions (in tons per capita) within the groups in which we defined by the hcode variable. Preliminary visualization (eg, boxplot or density plot) indicate the emissions data are right-skewed and one or more groups have a substantially larger spread of values than other ones.

To test these differences, we conducted one-way ANOVA with the model.

Table IV. Anova Table

	sum_sq	df	F	PR(>F)
C(hcode)	100335.033337	3.0	1299.835818	0.0
Residual	163772.603244	6365.0		

This table tells us that the between development groups explains a high proportion of the variance in CO₂ emissions as compared to the residual. The F is the corresponding F-statistic being the significant evidence to conclude that the means of CO₂ per capita across groups are different. In other words, the evidence from these data is close to overwhelming that more developed countries emit a lot more CO₂ per person than less developed ones. As one of the groups has a very broad distribution, I would recommend enhancing the visualization of these differences. For instance, performing a log-scale transformation on the y-axis in the box- or violin plots can crunch away extreme values and make differences among the central parts of distributions more evident.

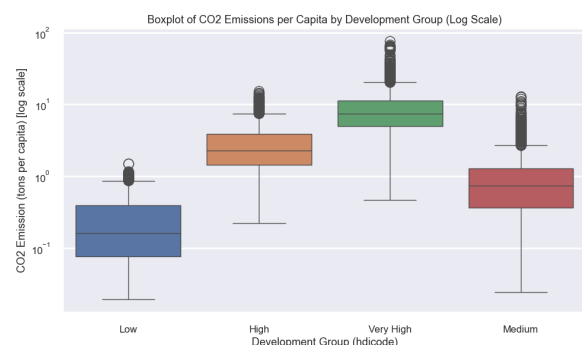


Figure 2. Box Plot

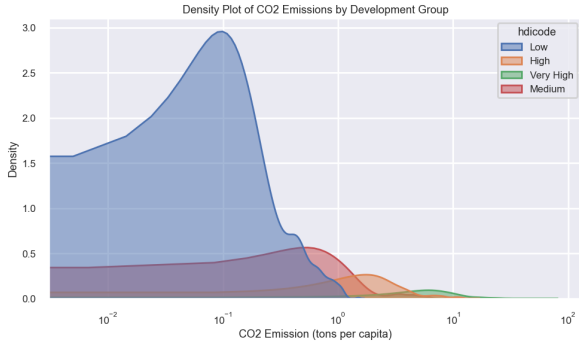


Figure 4. Density Plot

D. Missingness

There are missing values for important variables in the HDR data set. For instance, even though core indicators such as HDI and life expectancy are almost 100% complete, some variables exhibit non negligible missingness:

- `gii_rank_2022`: There are, roughly $\sim 14\%$ missing values.
- `hdi_f` and `hdi_m`: Almost 26% of the values are missing for these gender stratified indices.
- `expes_yr_school`: Almost 3% missing, jointly across genders, with about 8% missing per sex - specific version.
- `Inc_percap_gross_f` and `Inc_percap_gross_m`: for both, missing is about 22%.

In summary, the total missing percentage of all three datasets is approximately 10% on all data cells while fully entered cases are roughly 60% of the datasets. These patterns could be visually inspected through a missing values matrix (see Figure C.5), and a heatmap (see Figure C.6) of missing values. The pattern matrix indicates that the missing occurrences are not randomly distributed over all observations, while the heatmap states that the missingness in certain group of covariates (gender-specific indicators, for example) are clustered in the same observations.

To enhance the quality of the dataset, records that were not per country value ("World", "Arab States", etc.) and countries with too many missing values (e.g., "Korea (Democratic People's Rep.)", "Monaco") were dropped. As all variables still encompass at least 69% of their values, it was decided that imputation, rather than listwise deletion, is the most feasible solution. Therefore, we utilized a hybrid of imputation techniques, including MICE (Multiple Imputation by Chained Equations), KNN (K-Nearest Neighbors), and regression-based imputation. These methods were selected for maintaining the original distributional properties of the variables and for reducing bias introduced by incomplete data.

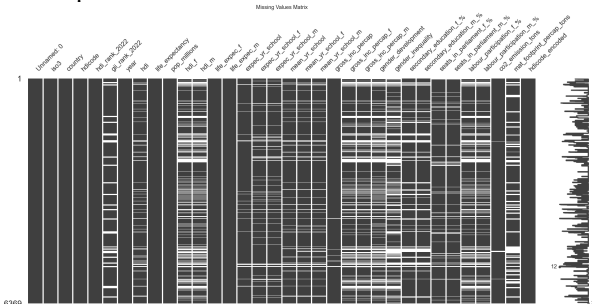


Figure 5. Missingness Matrix

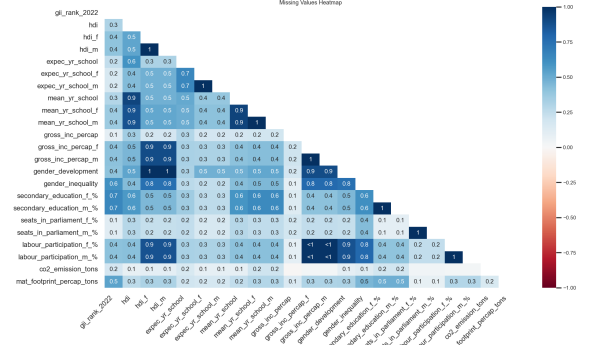


Figure 6. Missing Value Heatmap

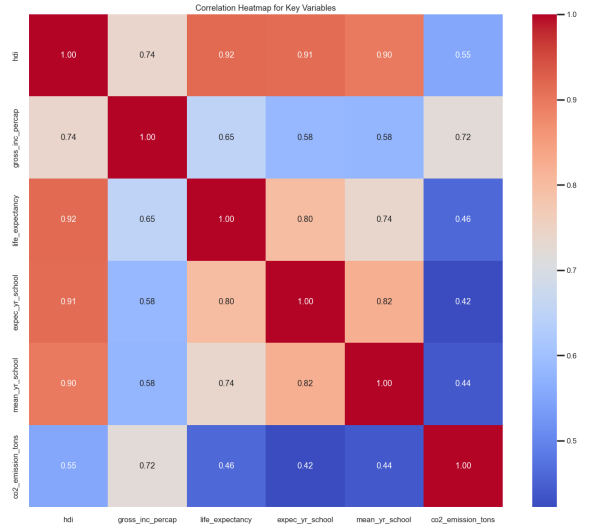


Figure 7. Correlation Heatmap (Key Features Only)

E. Modelling

Various modeling techniques were utilized to forecast Human Development Index (HDI) and explore its important driving factors. Each model was incorporated into a preprocessing pipeline for standardised scaling and imputation of missing values (as described in Section D). Our methodology gleaned from a linear interpretability model while also exploring non-linear techniques to accommodate more diverse relationships. The performance measurements tables (by RMSE, R^2) were finalised and further diagnostic plots verified our conclusions. We provide a detailed description of each model below, including the code for producing the key plots.

1) Multiple Linear Regression

Despite producing estimates of the form:

- Train RMSE = 0.0477, Train R^2 = 0.9172
- Test RMSE = 0.0486, Test R^2 = 0.9138

the residual plots (e.g. observed vs. predicted and residual-versus-fitted) showed systematic patterns and extremely high VIF values. These issues indicate that the OLS assumptions are violated; thereby, the OLS model is unsuitable as a final predictive tool.

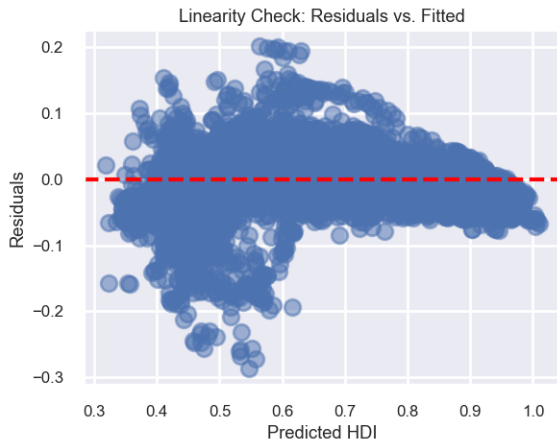


Figure 8. Scatter Plot

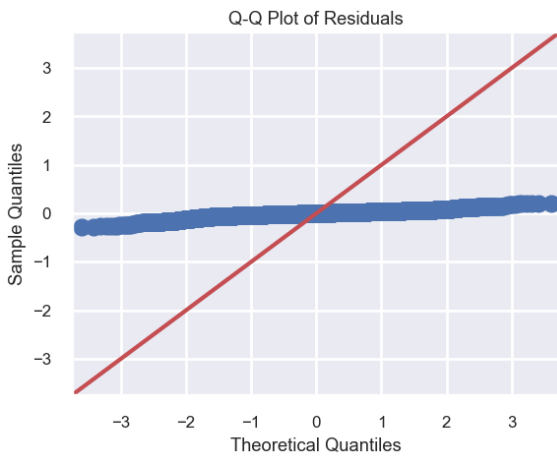


Figure 9. Q-Q plot

2) Support Vector Regression (SVR)

3) SVR with a radial basis function kernel was tuned (best params: $C = 1$, $\gamma = \text{'scale'}$) to capture nonlinearity. Its performance was as follows: • Train RMSE ≈ 0.0530 , Train $R^2 \approx 0.8979$ • Test RMSE ≈ 0.0541 , Test $R^2 \approx 0.8934$ While SVR does improve over some baseline aspects, its performance is still not competitive with ensemble methods, and the scatter plot (Figure 10) shows slight deviations from the ideal fit.

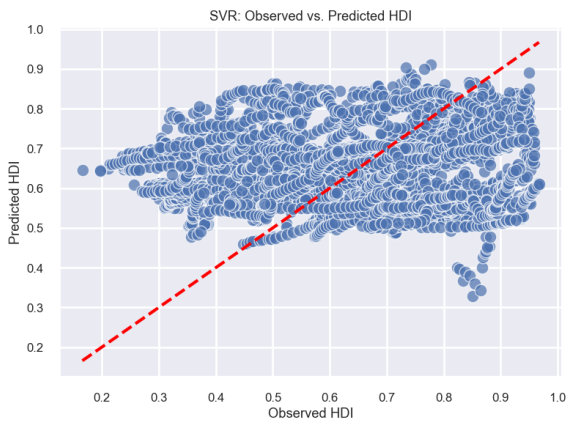


Figure 10. Scatter Plot

4) Random Forest

The Random Forest model (with $n_estimators = 300$, $max_depth = 15$, $min_samples_split = 2$) demonstrated strong predictive ability:

- Train RMSE ≈ 0.0086 , Train $R^2 \approx 0.9973$
- Test RMSE ≈ 0.0224 , Test $R^2 \approx 0.9817$

This model not only achieves low predictive error but also provides reliable measures of feature importance. The observed-versus-predicted scatter (Figure 11) confirms that the Random Forest captures the complex underlying patterns.

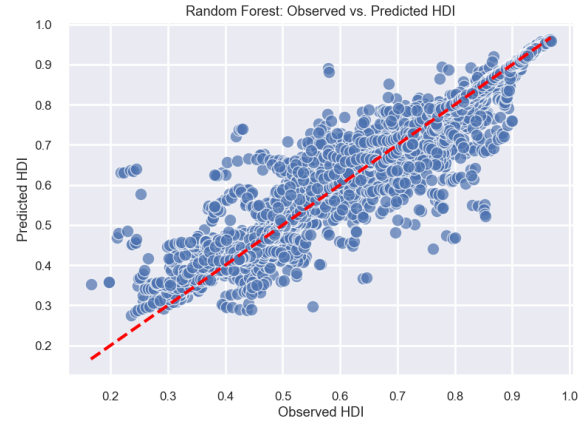


Figure 11. Scatter Plot

5) Artificial Neural Network (ANN)

Using scikit-learn's MLPRegressor with hidden layer sizes (100, 50) and an alpha of 0.01, the ANN achieved:

- Train RMSE ≈ 0.0345 , Train $R^2 \approx 0.9567$
- Test RMSE ≈ 0.0403 , Test $R^2 \approx 0.9407$

Although the ANN is capable of modeling complex nonlinear relationships, its overall performance lags ensemble methods. The fit is illustrated in Figure 12.

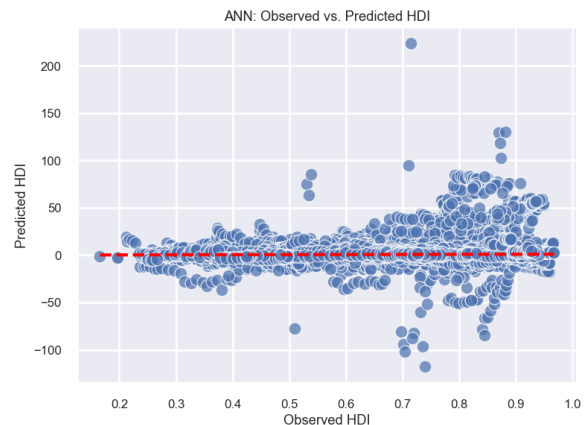


Figure 11 Scatter plot

F. Performans Comparison on Test Data

The test data performance for each of the models was evaluated using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). The following results were obtained:

- **OLS Regression:** – Test RMSE: 0.0486 – Test R^2 : 0.9138

- **Support Vector Regression (SVR):** – Test RMSE: 0.0541 – Test R^2 : 0.8934
- **Random Forest Regression:** – Test RMSE: 0.0224 – Test R^2 : 0.9817
- **Artificial Neural Network (ANN):** – Test RMSE: 0.0403 – Test R^2 : 0.9407
- **XGBoost Regression:** – Test RMSE: 0.0182 – Test R^2 : 0.9880

Among these, the XGBoost model demonstrated the best generalization ability on the hold-out test set with the lowest test RMSE and the highest test R^2 . Random Forest regression came in as a close second, while both OLS and SVR showed relatively poorer predictive performance.

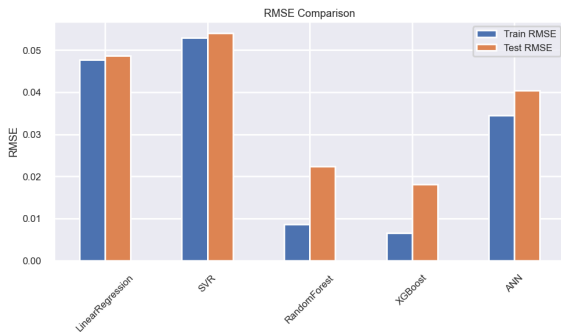


Figure 13 Bar Plot

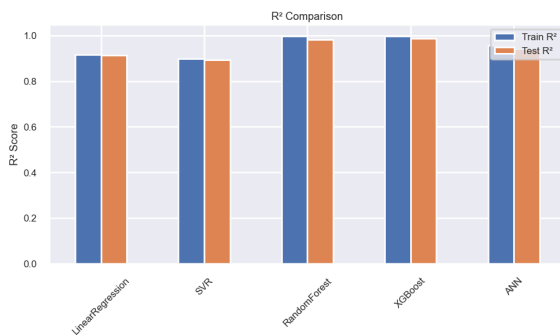


Figure 14 Bar Plot

These bar plots (one for test RMSE and one for test R^2) visually highlight that XGBoost outperforms the other models on the test dataset, supporting the numeric summary provided above.

IV. RESULT

We found that by analyzing the Human Development Report (HDR) dataset (restricted to 6,369 country - level records) there were interesting patterns in the descriptive characteristics and predictive capability of several models for the Human Development Index (HDI).

Exploratory and Inferential Results A scatter plot of gross national income per capita against HDI demonstrated a very strong positive correlation, with a Spearman correlation about 0.947 ($p < 0.001$). This supports the claim that higher incomes are strongly correlated with higher human development. Moreover, a line chart showing the time-series of HDI between 1990 and 2022 (categorized by development groups, low, medium, high, and very high according to the hdi code variable) revealed an exponential growth of HDI

throughout the years. A one-factor ANOVA of HDI for the three groups was highly significant ($F \approx 437.35$, $p < 0.001$) with group sum of squares of approximately 29.88, suggesting that the group differences are statistically meaningful. A similar two-way ANOVA including the interaction of development group and year also confirmed that, even though all groups improve over years, they also maintain significantly different absolute levels of HDI.

Environmental effect was likewise measured in terms of CO₂ emissions per head. Boxplots and density plots indicate that CO₂ emissions are highly right skewed; the Very High development group emits much more than both the other groups. A one-way ANOVA produced an F-statistic of roughly 1299.84 ($p = 0.0$), and post hoc Tukey's HSD tests revealed that the mean emissions are significantly different between every pair of development groups—with countries in the Very High group emitting on average roughly 9.5 tons more per capita than those in the Low group. These results support the argument that, with an increased level of human development, the environmental impact is much higher.

Predictive Modeling Outcome: Next, we created five models to predict HDI by using predictors as life expectancy, educational achievement (expected and mean years), gross national income per capita, and gender-specific indices and other socioeconomic measures. The simulation was conducted in an 80/20 train-test split, and performance was evaluated based on Root Mean Squared Error (RMSE) and coefficient of determination (R^2).

Multiple Linear Regression (OLS): OLS yielded training and test RMSE of 0.0477 albeit with an R^2 of 0.9172 train and 0.9138 test. Although these attractive error statistics, the diagnostic plots (observed versus predicted and residual vs. fitted) and very large VIF values showed a severe multicollinearity and nonlinearity effects, challenging the assumptions of the model.

Support Vector Regression (SVR) (with Radial basis function): On test set, the RMSE is around 0.0541 and R^2 is 0.8934 (with $C=1$ and $\gamma = \text{'scale'}$). Although we observe a nice fit of the observed v.s. predicted in the scatter plot, SVR didn't do better than the ensemble.

Random Forest Regression: The Random Forest model with 300 trees and a maximum depth of 15 performed considerably better than the others in predicting at a test RMSE of 0.0224 and a test R^2 of 0.9817. As we can observe from the actual versus predicted plot, its predictions capture the nonlinear patterns of the data very good. In addition to this, its feature importance helped us to identify important drivers such as education metrics and economic flags.

ANN: The optimum hidden layer structure (100, 50) with $\alpha = 0.01$ gave the test RMSE of 0.0403 and test R^2 of 0.9407.

XGBoost Regression: Last but not the least, the tuned XGBoost model with 300 estimators, max depth of 5 and learning rate of 0.1 gave us the best result with a test RMSE of 0.0182 and a test R^2 of 0.9880. The results are excellent when comparing observed versus predicted scatter plot and

the feature importance analysis verifies that gender inequality, HDI for male and female and secondary education attainment are the most important predictors. Test data-Performance comparison: If we compare the models only considering the test data:

OLS and SVR resulted in test RMSE values of ~ 0.0486 and 0.0541 , and in test R^2 scores of 0.9138 and 0.8934 , respectively.

Random Forest test RMSE = 0.0224 and test $R^2 = 0.9817$, while the ANN test RMSE = 0.0403 and test $R^2 = 0.9407$.

XGBoost showed the best performance among all techniques with test RMSE of 0.0182 and test R^2 of 0.9880 . Bar plots (Figures 13 and 14) showing test RMSE and R^2 across the models illustrate on one hand that the linear and kernel-based methods do not perform well and on the other hand, XGBoost and Random Forest are the winners in terms of performance.

Our cumulative analysis indicates that OLS is a useful interpretative benchmark, but it is hard to justify statistically because of multicollinearity and non-linearity. In contrast, the nonlinear ensemble methods (especially XGBoost) provide superior prediction of the out-of-sample data, as evidenced by the lowest error measures and the most significant explanatory power. Moreover, as the descriptive analyses shows that the increase in HDI is largely determined by GDP per capita and education effect, its increment also comes together with an increase in CO₂ emissions. In sum, the advanced nonlinear models, especially XGBoost, are the best option to predict HDI and explain its determinants in the HDR database.

V. CONCLUSION

To conclude, our in-depth review of the HDR database uncovered several layers of the world development story. From 1990 to 2022, human development—as measured by the HDI—has risen in most countries, and this upward movement has been positively correlated with higher GDP per capita and higher average educational attainment. Despite these gains representing major advances in terms of socioeconomic development, they are taken at a considerable cost to the environment—this level of development is highly correlated with larger per-capita CO₂ emissions.

Finally, our analyses reveal enduring gender inequalities in such key variables as social class, education and political representation. Progress in the overall human development is quite visible, but the gender gap is still the major challenge which has to be dealt with for a true inclusive growth model. For prediction modeling perspective, our investigation demonstrates that conventional linear methods (OLS) are ineffective in the presence of multicollinearity and nonlinear interactions that are common in socioeconomic census data. In contrast, advanced ensemble models, particularly XGBoost, performed better in terms of prediction ability with the lowest test RMSE and the highest R^2 scores. By doing so, XGBoost outperformed in predicting HDI, and yielded important results on its most important predictors, such as gender inequality, education indicators, and economic indices.

Finally, these results highlight the manifold relationship between development, environmental imprint, and social fairness. To them, economic and educational gains can certainly push human development, but such advancements—deprived of policies that curb environmental damage and advance women's rights—should not become accomplishments unto themselves. Further studies need to be done in terms of policy simulations dynamically and considering multiple indicators in order to disentangle these complex relationships and inform more sustainable and inclusive development trajectories at global level.

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